Experiment 7: Write the python program Breadth-First Search (BFS)

Aim:

To write a Python program to implement Breadth-First Search (BFS) algorithm.

Algorithm:

1. Start from the initial node.
2. Add the node to the queue.
3. Visit and print the first node in the queue.
4. Add its unvisited adjacent nodes to the queue.
5. Repeat until the queue is empty.

Program:

```
graph = {
   'A': ['B', 'C'],
   'B': ['D', 'E'],
   'C': ['F'],
   'D': [],
   'E': [],
   'F': []
}


visited = []
queue = ['A']


while queue:
   node = queue.pop(0)


   if node not in visited:
      print(node, end=" ")
      visited.append(node)
```

queue.extend(graph[node])

exp 7 BFS.py - C:/Users/yadal/OneDrive/Documents/AI/exp 7 BFS.py (3.14.7)

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```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

visited = []
queue = ['A']

while queue:
    node = queue.pop(0)

    if node not in visited:
        print(node, end=" ")
        visited.append(node)
        queue.extend(graph[node])
```

Output:

```
>>
========== RESTART: C:/Users/yadal/OneDrive/Documents/AI/exp 7 BFS.py ==========
A B C D E F
>>
```

Result:

Thus, the Python program to implement Breadth-First Search (BFS) was executed successfully.